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RESEARCH ARTICLE

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Reporting gaps between news media and scientific papers on outdoor air pollution–related health outcomes: A content analysis

Zhuanlan Sun[1] | Jun Cang[1] | Yuhui Ruan[2] | Demi Zhu[2]

[1]Department of Management Science and Engineering, Tongji University, Shanghai, China

[2]Department of Comparative Politics, Shanghai Jiaotong University, Shanghai, China

Correspondence
Demi Zhu, Department of Comparative Politics, Shanghai Jiaotong University, 1954 Huashan Road, Shanghai 200030, China.
Email: zhudemi@sjtu.edu.cn

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Summary

Objectives: In China, news media are useful for educating the public about the health threats of air pollution. To explore the potential gaps between scientific findings and the public's understanding of them, the characteristics of news media articles and their corresponding scientific papers were analysed.

Methods: We used 22 articles relating to the health outcomes of exposure to outdoor air pollution published on *Baidu News* over the past year. An assessment tool developed by Robinson et al was used to evaluate the quality scores of news articles. Pearson correlation coefficients were used to measure the relationship between news media reporting and the characteristics of scientific papers. Misleading reporting, interpretation, or extrapolation in headlines and text bodies of news articles were examined.

Results: The quality scores of the news articles ranged from −4 to 8, with an overall median score of 3. Correlation results showed that the scientific papers citation in Twitter ($r = .88$, $P < .001$) and Facebook ($r = .64$, $P < .01$) were significantly and positively associated with their citations in news stories. Media misunderstanding of scientific findings was common: 15 news headlines were identified with at least one spin (misrepresentation of scientific results), and 12 news articles had seven types of spin in the body texts.

Conclusion: Little media attention has been paid to scientific findings by Chinese researchers. Therefore, researchers and science journalists in China should make a better effort

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to engage in accurate and informative public discourse on domestic research.

KEYWORDS
content analysis, health outcomes, news media, outdoor air pollution, scientific papers

1 | INTRODUCTION

Outdoor air pollution poses a major health risk. This has been well established for decades.[1] It is estimated that 4.2 million premature deaths were caused globally by outdoor air pollutants in 2016,[2] and the figure has increased continuously. The quantitative relationship between outdoor air pollution and health outcomes has been well documented in a wide variety of scientific papers, with those health outcomes including cardiovascular disease,[3] respiratory disease,[4] lung cancer,[5] and mortality,[6] among others.[7-9] However, a general understanding of relevant scientific researches and accurately communication of research findings to the public is limited.

At present, news media is more influential, informing and educating the public that air pollution has a negative effect on health.[10] Scientific results reported in news media can influence public health perceptions and individual behaviour.[11,12] These media include magazines, newspapers, blogs, and social media.

The majority of news coverage on outdoor air pollution–related health outcomes originates in the scientific researches. Thus, the dissemination of scientific research in an unbiased, undistorted, and efficient manner is vital to both the public and the scientific community.[13] However, media misunderstanding of scientific findings is common in science news reporting.[14,15] The media's interest lies in speedy and eye-catching publication rather than accurately reporting of scientific information, which gives rise to public beliefs about potential misinformation not grounded in scientific evidence.[16] This intentional or unintentional misrepresentation, exaggeration, and oversimplified reporting of scientific research results have induced an inaccurate public understanding of scientific findings and have negatively impacted the public view on socio-scientific issues.[17-19]

As a consequence, the relationship between science and public understanding has been characterised by terms such as “distance” and “gap.”[20] There are sometimes striking differences between scientists' and the general public's understanding of scientific results.[21,22] To understand the extent and nature of this “gap,” we analysed the characteristics of texts on outdoor air pollution and its health outcomes obtained from Chinese news articles and corresponding scientific papers. Our aim is to provide an insight into how online news articles portray findings on health effects in scientific papers and to explore the mechanism of misleading reporting, so that it can improve the public's understanding of the health risks associated with air pollution and journalists and researchers can find ways to improve their communication to the public.

2 | METHODS

2.1 | Selection of news articles and scientific papers

Because of the decline in the popularity of traditional print-based newspapers and the increasing favour for digitised media,[23] we searched for news articles that cited scientific papers on the health outcomes of exposure to outdoor air pollution during the past year (2018.05.26–2019.05.26) on *Baidu News*. *Baidu News*, launched in 2003, is one of the largest and most up-to-date online news services in China and attracts billions of users a week to its news

content. It consolidates news from more than 500 authoritative websites and includes both domestic and international information.

Each search began with consideration of specific outdoor air pollutants ("air pollution," "$PM_{2.5}$," "PM_{10}," "sulphur dioxide," "nitrogen dioxide," "O_3," "nitrogen oxides," "toxic air pollutants," OR "volatile organic pollutants") combined with unspecific terms ("research"). All the search terms were in Chinese. News articles that referred to the results of one or more scientific papers were included. We excluded news articles that reported (a) results not relevant to scientific papers; (b) results that originated in posters, reports, or blogs about scientific research; and (c) results from a scientific paper that had already been reported by previous article in our sample.

The references in the extracted news articles were used to conduct literature retrieval of the corresponding scientific papers. If a selected news article did not refer to scientific papers, other information relating to the scientific papers, such as the authors, the scientific journals, and the combination of keywords relevant to the research findings, was used to run a systematic search using Google or Google Scholar.

2.2 | Information extraction from news articles and scientific papers

We extracted the following information from the news articles and their corresponding scientific papers: (a) country, (b) journal of publication, (c) publication date, (d) study population, (e) sample size, (f) study design, (g) study limitations, and (h) altmetric scores representing the social media attention garnered by the scientific papers.[24] Altmetrics, first proposed by Priem et al,[25] is a way to understand the potential impact of scientific research on the public and the scientific community.[26] It analyses scientific publications in terms of online discussion around the research results in social networks, news outlets, policy documents, blogs, and other scholarly and nonscholarly platforms. It includes citations on social networks such as Twitter and Facebook and in news outlets, which are not available in Google Scholar and the Web of Science.[27] Altmetrics scores are major indicators for identifying publications with the potential for high citation in the future.[24] In this study, the R package "rAltmetric" was employed to retrieve the altmetric scores of the selected scientific papers.[28] The generated altmetric scores included the altmetric news score, altmetric Twitter score, altmetric Facebook score, and so on, which are characterised by the scientific papers' citations in different online sources. Information on the news articles and their corresponding scientific papers was extracted by one author of this study and verified by another.

2.3 | Quality assessment of news articles

An assessment tool, developed by Robinson et al,[29] was used to evaluate the reporting quality of news articles. This tool is designed specifically for online health news and has been applied by other recent studies.[30] The tool consists of 21 items and assesses various aspects of a news article's quality, including the credibility of the sources, the contexts of the studies, and the representativeness of the research used. Quality scores calculated by this tool are awarded from a baseline of zero, and articles can receive maximum and minimum quality scores of 17 and −12, respectively. The descriptive statistics used for the quality score are median values and the interquartile range (IQR).

2.4 | Correlation analysis

Pearson correlation coefficients were used to measure the relationship between news citation score and the characteristics of scientific papers. The news citation score is the counts of news stories in different news outlets that cite a particular scientific paper. The characteristics of scientific papers are represented in two parts. For the news media factors, the news attention given to a scientific paper on Twitter and Facebook was measured using the respective altmetric scores. For the factors relevant to scientific papers, the impact factor data of the journals in which the papers were published was collected from the Journal Citation Reports (JCR) in 2018, the citation counts were collected from Google Scholar data, and the article quality was represented by the quality scores obtained using

assessment tool of Robinson et al, as introduced in Section 2.3. By conducting a correlation analysis, we sought to analyse the potential factors that significantly influence the citation outcomes of news articles.

2.5 | Interpretation of scientific papers

On the basis of the classification by Haneef et al,[18] we assessed whether there was any misleading reporting, interpretation, or extrapolation in (a) headlines and (b) the text bodies of these news articles and labelled the misrepresented results the "spin."[31]

3 | RESULTS

Figure 1 displays the systematic search process used to isolate news articles and scientific papers relevant to this study. In total, 263 *Baidu* online news articles met the search criteria. Of those, 44 were selected for in-depth screening after the application of the inclusion and exclusion criteria set at the beginning of the news article identification process. Finally, 22 articles were used for assessment and analysis after a full-text screening. Twenty-five scientific papers were referenced by the final 22 news articles, of which two news articles cited two and three scientific papers, respectively.

3.1 | Analysis of the characteristics of the selected news articles and scientific papers

A list of news articles with referenced scientific papers included in the analysis is given in Table S1. With regard to the characteristics of the news articles listed on the left side of Table 1, 20 (90.9%) news articles were reported by a general news outlet, while only one news article is from a scientific and biological news outlet. All the news articles in our sample reported on the study population; the sample size was reported in 90.9% of news articles, and the study design was reported in 13.6% of them. In contrast, only one news article reported the study limitations, and two news articles contained an electronic link or reference to the scientific papers. The news articles varied in terms of the presence of the reporter's name: Half of the selected news articles contained the reporter's name, and 50% of

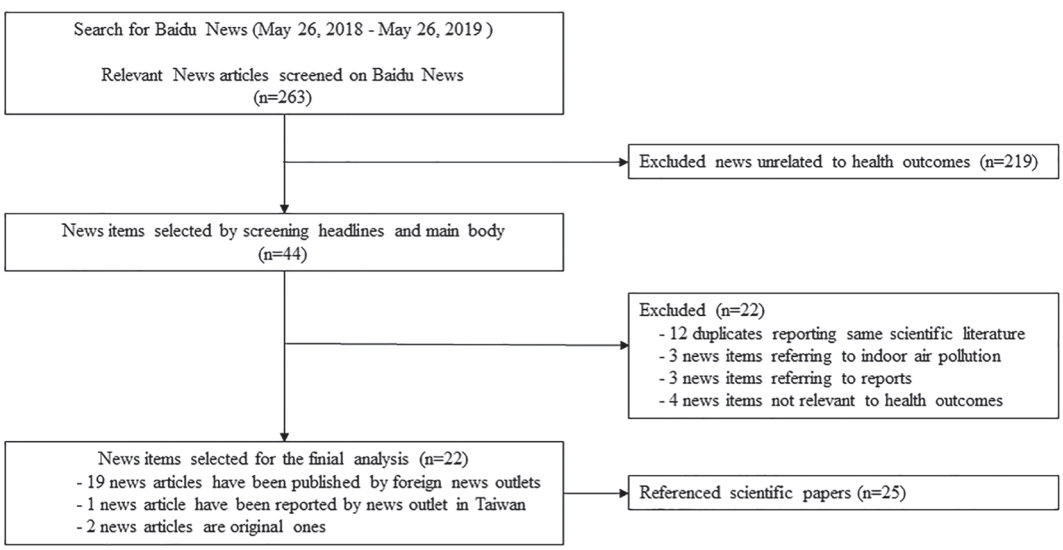

FIGURE 1 Descriptions of general characteristics of the selected news articles and scientific papers

TABLE 1 General characteristics of the included news articles and scientific papers

News Articles Characteristics (n = 22)		Scientific Papers Characteristics (n = 25)	
Type of news outlet, n (%)		**Type of news outlet, n (%)**	
– General news outlet	20 (90.9)	– Medical science	15 (60.0)
– Scientific news outlet	1 (4.5)	– Environmental science	4 (16.0)
– Biological news outlet	1 (4.5)	– Multidiscipline	6 (24.0)
Reporting items of		Study population, n (%)	
– Study population, n (%)	22 (100.0)	– Human	23 (92.0)
– Sample size, n (%)	20 (90.9)	– Animal	2 (8.0)
– Study design, n (%)	3 (13.6)	Study design, n (%)	
– Study limitations, n (%)	1 (4.5)	– Cohort [2,3,4,6,10,12,15-17,19-20,22][a]	12 (48.0)
– Link or reference to scientific paper, n (%)	2 (9.1)	– Cross-sectional [4,11]	2 (8.0)
– Reporter's name, n (%)	11 (50.0)	– Other	13 (52.0)
Top two population (if human study), n (%)		Health outcomes, n (%)	
– Children & adolescent	10 (50.0)	– Mental disorder	10 (40.0)
– Elderly	4 (20.0)	– Cardiovascular diseases	5 (20.0)
Published by foreign news, n (%)		– Hospital related	5 (20.0)
– British news outlets	6 (30.0)	– Respiratory diseases	3 (12.0)
– American news outlets	1 (5.0)	– Chronic diseases	2 (8.0)
– Other unspecified foreign news outlets	13 (65.0)	Sample size (range)	(145-73,000,000)

[a]Serial numbers of the news articles can be found in Table S1.

them were anonymous. Of the 20 news articles on human studies, the largest surveyed population was under the age of 20 (10/20, 50%), followed by a group of elderly at 20% (4/20).

Ninety percent of the news articles had been published by foreign news outlets. Six of the included articles were translated from British news outlets, and one had been published by a news outlet in America; the remainder were published in other foreign countries and did not contain detailed information of the news outlets.

Of the 25 scientific papers, 15 (60%) were published in medical journals, four (16%) in environmental journals, and six (24%) in multidisciplinary journals (right side of Table 1). The study of a human population was done in 23 articles, and two scientific papers reported animal-based research. Primary research studies were favoured by the news reports; the dominant study design used in these studies was cohort, with a corresponding number of 12. The sample sizes of the scientific papers varied widely, from more than one hundred to millions (range: 145-73 000 000).

Nearly all of the news articles were reported in a timely manner. Following the health outcome classification in previous research,[32] the health outcomes reported frequently were mental disorders (10/25, 40%), cardiovascular diseases (5/25, 20%), hospital-related problems (5/25, 20%), respiratory diseases (3/25, 12%), and chronic diseases (2/25, 8%). The journals in which the scientific papers had been published varied in terms of impact factor, with prominent journals (such as JAMA, Lancet, and PNAS) also on the list.

3.2 | Quality assessment of news articles

The quality scores of the news articles ranged from −4 to 8, with an overall median score of 3 (Table 2). Based on the article quality categorisation recommended by Robinson et al,[29] one in five news articles scored less than 0 and was categorised as poor quality, and 80% scored 0 to 10 and were categorized as satisfactory quality; no high-quality news article scored more than 10 in the included samples.

Considering that the scores evaluated using the quality assessment tool were not normally distributed and most news articles were reported by foreign news outlets, we categorised the articles into four groups and classified the scores into two groups. The actual quality score distribution is presented in Figure 2 and Table 2: Median quality scores for the United States and the United Kingdom news outlets were both equal to 3, and those for news outlets in China and other unspecified foreign countries were 4 and 5, respectively. The percentage of news articles that achieved a score between 0 and 10 (thus categorised as satisfactory quality) varied for news articles. Articles with the lowest and highest scores fell under the groups of the US news outlets (1/25, 4.0%) and other unspecified foreign news outlets (11/25, 44.0%), respectively. Unspecified foreign news outlets accounted for the highest number of poor-quality news articles (16.0%).

3.3 | Correlation analysis

Given that similar scientific papers were published in the same journals as the sample papers but did not generate news coverage, further analysis was conducted to identify which factors might have an influence on news reporting.

TABLE 2 Descriptive analysis of quality scores

		Quality Score		Quality Category	
News Articles	n (%)	Median Score	IQR Range	Poor, n (%)	Satisfactory, n (%)
The US news outlets	1 (4)	3	3 to 3	0 (0)	1 (4.0)
The UK news outlets	6 (24)	3	2.25 to 3.75	1 (4.0)	5 (20.0)
Other unspecified foreign news outlets	15 (60)	5	-0.25 to 6	4 (16.0)	11 (44.0)
China's news outlets	3 (182)	4	3.5 to 6	0 (0)	3 (12.0)

Abbreviation: IQR, interquartile range.

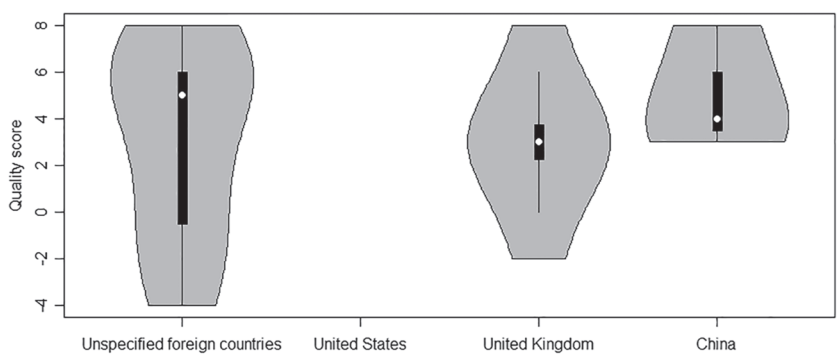

FIGURE 2 Violin plot of quality score for different countries' news outlets

The correlation result of the news citation score with the media factor and scientific paper factors is presented in Figure 3 and Table 3. News attention from social media, including Twitter (r = .88, P < .001) and Facebook (r = .64, P < .01), was significantly and positively associated with citations in the news; however, none of the scientific paper factors showed significant correlation with news citation.

3.4 | Interpretation of scientific research

From among the 22 news articles, 15 news headlines were identified with at least one spin (Table 4). Headlines of news articles that reported studies on animals, such as mice, always indicated overgeneralisation and extrapolated the results to human application (n = 2). Among news articles reporting studies on humans (n = 20), headlines with an implied linguistic spin were more pronounced (n = 8). For example, a headline reporting a study evaluating the impact of air pollution exposure on cognitive performance stated "Air pollution makes people *retarded*" (translated from Chinese). Another example was a headline using the word "culprit," while the scientific paper estimated the global burden of air pollution on asthma. For news articles reporting an observational study design (n = 22), 13.6% (3/22) of headlines claimed causal statements and 18.2% (4/22) presented exaggerated statements. For instance, when reporting on scientific research that only found evidence of a positive association between air pollution and incidence of dementia, the headline of a news article used causal language with the word "cause." The headline, "Some London parents have decided to leave the polluted city for the sake of their children's health," reported on a study that only found that smaller lung volume in children was associated with higher air pollution exposures.

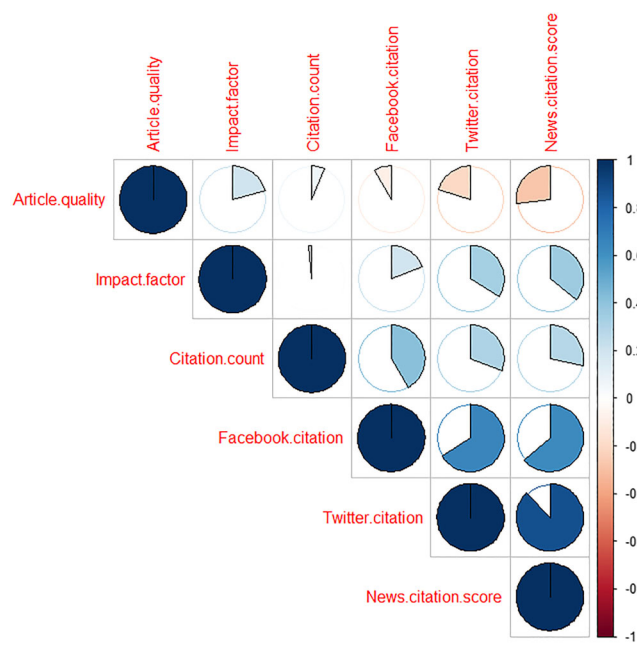

FIGURE 3 Correlation plot of factors related to the news citation score

TABLE 3 Correlation of the news citation score with the media and scientific paper factors

	Media Factor		Scientific Paper Factor		
	Twitter Citation	Facebook Citation	Impact Factor	Citation Count	Article Quality
News citation score	0.88***	0.64**	0.35	0.28	−0.27

**P < .01.
***P < .001.

TABLE 4 Spin in news articles

Spin in Headlines (n = 15)	n (%)	Spin in Text (n=12)	n (%)
Misleading reporting		Misleading reporting	
– Selective reporting [2][a]	1 (4.5)	– Selective reporting	1 (4.5)
– Linguistic spin [5,12-17,19]	8 (36.4)	– Linguistic spin	2 (9.1)
Misleading interpretation		Misleading interpretation	
– Exaggeration [9,11,17,22]	4 (18.2)	– Exaggeration	3 (13.6)
– Causal statements [7,16,19]	3 (13.6)	– Causal statements	2 (9.1)
Misleading extrapolation		Misleading extrapolation	
– Animal study results to human application [3,13]	2 (9.1)	– Animal study results to human application	2 (9.1)
		– Inappropriate picture	1 (4.5)
		– Extrapolating the effect of study outcome	1 (4.5)

[a]Serial numbers of the news articles can be referred to in Table S1.

We identified 12 news articles with seven types of spin in their body text (Tables 4 and 5). A total of three news articles contained misleading reporting. Linguistic spins were identified in two news articles, both of which stated "[…] pollution has remained stubbornly high." A third news article reported that various types of cardiovascular diseases were worsened by outdoor air pollution, which was only part of the information presented by the relevant scientific research.

TABLE 5 Misrepresentation in the text of 12 news articles

Spin Type	Examples	News Article (No)[a]
Misleading reporting		
– Selective reporting	Selective reporting of cardiovascular diseases	8
– Linguistic spin	"[...] Pollution has remained stubbornly high"	1,5
Misleading interpretation		
– Exaggeration	"[...] The risk of anxiety has increased"	2
– Causal statements	"Exposure to nitrous oxide during pregnancy was significantly associated with autism spectrum disorder"	12
	"British scientists have discovered an unexpected cause of the disease"	16
	"Air pollution is a major cause"	19
	"[...] Which is the reason of decrease in the total number of asthma cases recorded"	22
Misleading extrapolation		
– Animal study results to human application	Animal study of mice to human application	2,13
– Inappropriate picture	Pictures of air purifier in the main body	4
– Extrapolating the effect of study outcome	"[...] Students have to take crucial entrance exams on days with heavy pollution"	15

[a]Serial numbers of the news articles can be referred to in Table S1.

Regarding the misleading interpretation of scientific papers in the text of five news articles, three news articles used causal language, such as "a major cause" and "a leading factor," and two were identified as exaggerating the conclusion of the correlation between air pollution and health outcomes. For example, one news article reported, "Exposure to nitrous oxide during pregnancy was significantly associated with autism spectrum disorder," while the scientific research only detected an association between exposure to nitrous oxide and autism spectrum disorder.

Of the four news articles identified as having misleading extrapolation, two extrapolated the effect identified in an animal study of mice to humans. One news article used pictures of an air purifier in the main text, which reported the correlation between air pollution and children's academic performance. The final news article extrapolated the effect of the study outcome, reporting "students have to take crucial entrance exams on days with heavy pollution" in reference to a study evaluating the damage from air pollution exposure to intelligence.

4 | DISCUSSION

Clear reporting of scientific research outcomes in news media is important to help common readers understand the complex and challenging contents of scientific research. Our research focused on analysing news articles on health outcomes related to outdoor air pollution published on *Baidu News*, finding differences between these news articles and scientific papers they referenced, assessing the news articles' quality, and evaluating misrepresentation.

Although we retrieved national news articles using the largest Chinese search engine, only 22 relevant articles evaluating air pollution–related health outcomes were published globally within the past year. To our knowledge, studies on air pollution–related health outcomes are mainly conducted in the United States (168/799, 21.0%) and China (145/799, 18.1%).[32] Most news articles in this study (20/22, 91%) were reported by news outlets in foreign countries, which is inconsistent with the conclusion that local scientists receive more media attention in their own countries.[33] This nondomestic preference may be explained by factors such as availability of scientific literature and journal, which influence news outlets' coverage. Since most of this study's news articles were translated from other foreign-language newspapers, it seems that scientific findings published by Chinese researchers are not gaining the attention of domestic news journalists. Thus, more effort should be made to increase communication efficiency between news outlets and scientific communities in China.

As most of the included news articles were published in non-Chinese news outlets, we cannot observe a national bias. The distribution of studies reported in all news outlets did not mirror the global research output. The vast majority of reports were based on studies performed in the United States, as US scientists dominate in terms of the quantity of papers published.

We found that news articles were more inclined to report observational studies published in higher impact journals (one-third of the scientific papers were published in high-impact, prestigious journals, such as PNAS, Lancet, and JAMA). And other research designs, such as randomized controlled studies or meta-analysis, are less likely to be reported. Studies that were published in major journals with larger sample sizes were more likely to attract media coverage than those with smaller ones. Moreover, comparing the characteristics of scientific papers, scientific news coverage in social media was significantly and positively associated with citations in news outlets.

In news headlines that contain causal claims, causal interpretations are made for the scientific findings of observational studies, even though these studies only indicate correlations.[34-36] This leads to news articles presenting scientific results in an exaggerated manner.[36] On average, observational studies receive more attention,[37] and results of observational data can only be interpreted as risk analysis rather than causal effect detection. Thus, features of observational studies are often a key driver in increasing the general public's confusion about scientific research.[33] Our results show that correlated findings from animal studies are often extrapolated to human application.[19] The cumulative effect of exaggerated and misleading scientific reporting, even a subtle misrepresentation, might damage readers' understanding of health-related issues. Such misguiding information might influence people's health-related choices and result in inadequate policymaking. Training journalists in the accurate reportage of scientific research

and more effective communication between scientists and journalists should be encouraged to avoid confusion in reporting scientific results in news articles.

Our research inevitably has some limitations. The data collection was conducted over a limited period. The sample included only 22 news articles, which is not comprehensive and has the potential to lead some unknown bias. Moreover, our results cannot be generalised to other types of health-related news coverage on television and radio since we only analysed news articles. Future studies should therefore take into account for including more health-related topics, in order to examine further gaps between scientific research and public understanding of health issues. Finally, the quality assessment tool applied in our research is unique to health-related online news. Thus, tools for assessing the quality of online news sources are needed to measure articles using universally agreed standards.

5 | CONCLUSION

This study evaluated the quality of news reporting of scientific studies on outdoor air pollution and its health outcomes and explored the gap between the scientific researches and media reports. It is worth noting that Chinese scientific research has gained little attention from national news outlets. The results confirm that Chinese news articles on scientific findings were dominated by observational studies conducted in other countries. Thus, Chinese news outlets should endeavour to promote public discourse about domestic research on outdoor air pollution, which may help improve the public's awareness about scientific findings. Further, it is essential that researchers and science journalists make increased efforts to improve the quality of health news coverage and provide clear and informative news on public health without misrepresentation.

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CONFLICT OF INTEREST

The authors have no competing interests.

ANIMAL/HUMAN ETHICS

This article does not require any animal/human subjects to acquire such approval.

ORCID

Zhuanlan Sun https://orcid.org/0000-0003-4958-7351
Yuhui Ruan https://orcid.org/0000-0002-2583-4306

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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